

# The Data Design Processes

*DATA 3302: Data Visualization*

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*Professor Austin P. Wright*

Cal Poly San Luis Obispo · Fall 2026

Slides based on Munzner, Visualization Analysis and Design, 2014

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*Why involve direct representations?*

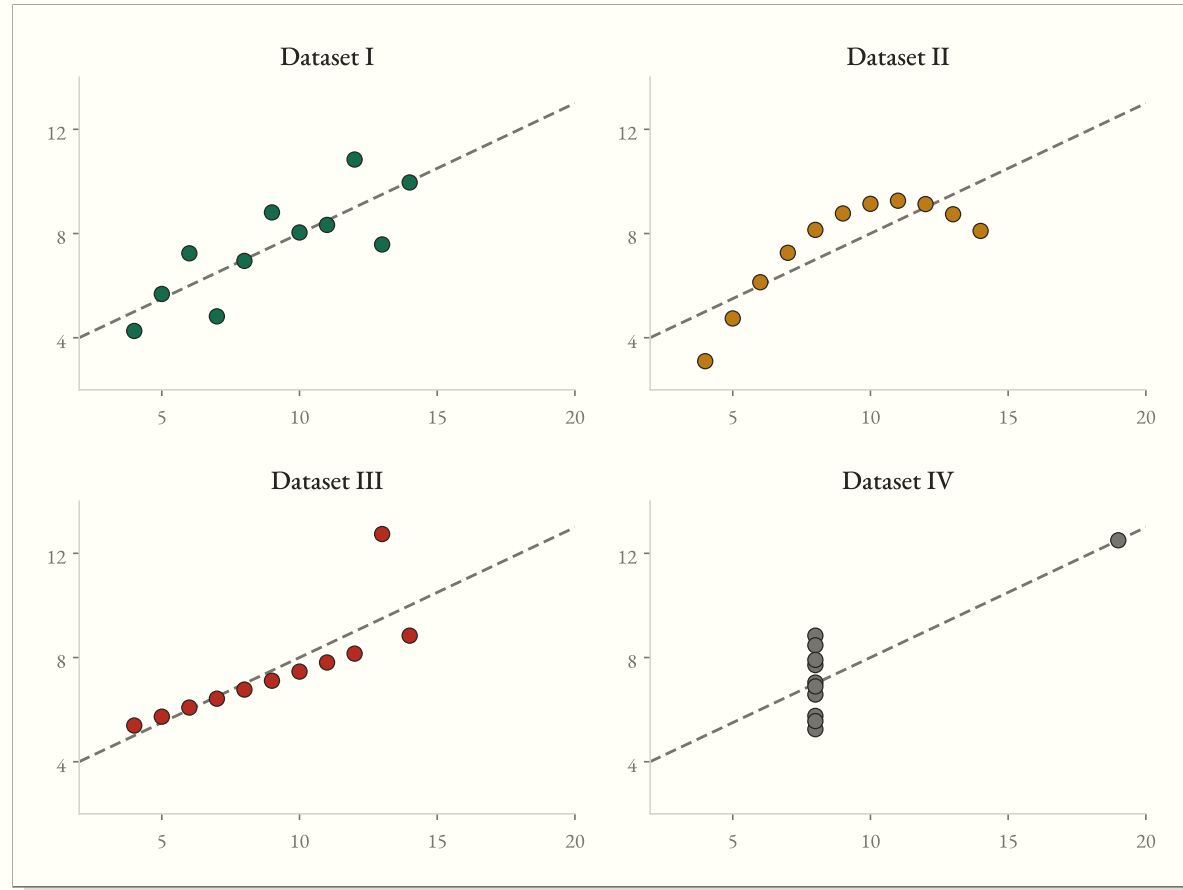
## *Anscombe's Quartet*

|                 | <i>I</i>       | <i>II</i>      | <i>III</i>     | <i>IV</i>      |
|-----------------|----------------|----------------|----------------|----------------|
| Mean of $x$     | 9.00           | 9.00           | 9.00           | 9.00           |
| Mean of $y$     | 7.50           | 7.50           | 7.50           | 7.50           |
| Variance of $x$ | 11.00          | 11.00          | 11.00          | 11.00          |
| Variance of $y$ | 4.13           | 4.13           | 4.12           | 4.12           |
| Correlation     | 0.82           | 0.82           | 0.82           | 0.82           |
| Regression Fit  | $y = 3 + 0.5x$ | $y = 3 + 0.5x$ | $y = 3 + 0.5x$ | $y = 3 + 0.5x$ |

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*Take 60 seconds:*

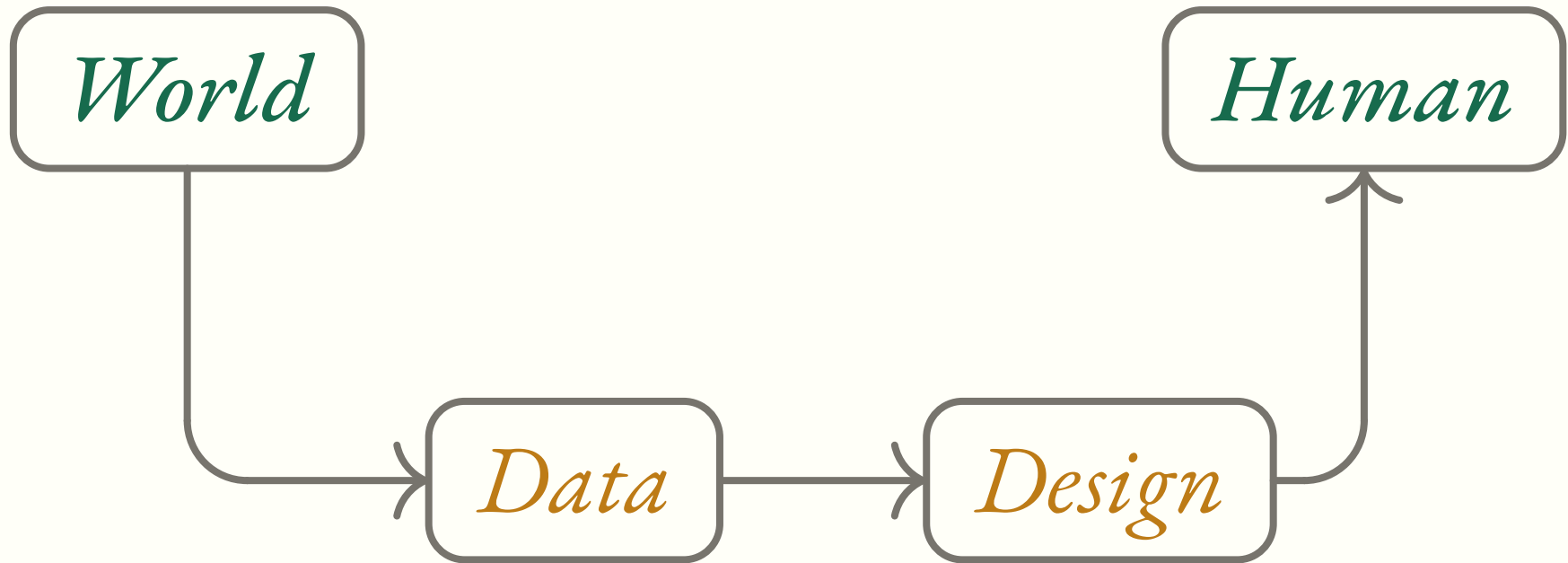
*All four datasets above share the same summary statistics. Sketch what you expect their scatterplots to look like.*



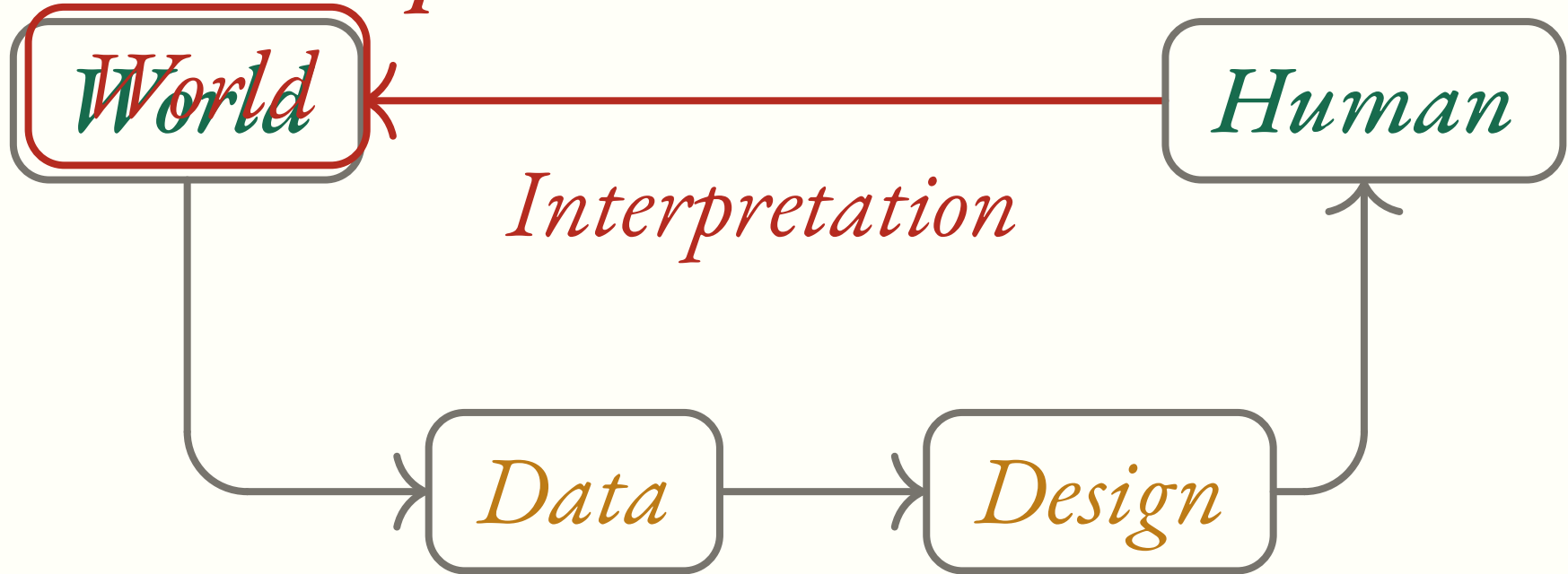
Anscombe, "Graphs in Statistical Analysis," 1973

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*Why is this difficult?*



*Mind the Gap!*



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*Discuss*

*Can you think of an example of a chart or statistic that led someone to an erroneous interpretation of the world?*

*Where in this loop did the gap open up?*

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# *Data Abstraction*

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## *Conceptual Model vs Data Model*

## *Data Types*

1. Items / Hard Index
2. Attributes / Variables
3. Links / Relations
4. Positions / Soft Index
5. Grids / Topologies

## *Data Types*

1. Items / Hard Index
2. Attributes / Variables
3. Links / Relations
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5. Grids / Topologies

## *Dataset Types*

1. Tables
2. Networks
3. Fields
4. Geometry
5. Sets

## *Dataset Availability*

1. Static / Offline
2. Dynamic / Online

## *Dataset Mutability*

1. Fixed / Immutable
2. Reactive / Mutable

# A City Bike-Share System

Riders check out *bikes* from *stations* scattered across downtown. Every *trip* links a start station to an end station. Each station reports its *GPS coordinates*.

Your task is to build a continuously updated citywide *demand heatmap*. You start by identifying the data abstractions.

## *Identify the Types*

1. Dataset Type?
2. Data Types?
3. Dataset Availability/Mutability?

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## *Attribute Types*

## *Attribute Types*

1. Categorical / Nominal
2. Ordinal
3. Quantitative (Interval)
4. Quantitative (Ratio)

## *Attribute Types*

1. Categorical / Nominal
2. Ordinal
3. Quantitative (Interval)
4. Quantitative (Ratio)

## *Order Directions*

1. Sequential
2. Diverging
3. Cyclic

| A        | B          | C               | S                 | T                   | U         |
|----------|------------|-----------------|-------------------|---------------------|-----------|
| Order ID | Order Date | Order Priority  | Product Container | Product Base Margin | Ship Date |
| 3        | 10/14/06   | 5-Low           | Large Box         | 0.8                 | 10/21/06  |
| 6        | 2/21/08    | 4-Not Specified | Small Pack        | 0.55                | 2/22/08   |
| 32       | 7/16/07    | 2-High          | Small Pack        | 0.79                | 7/17/07   |
| 32       | 7/16/07    | 2-High          | Jumbo Box         | 0.72                | 7/17/07   |
| 32       | 7/16/07    | 2-High          | Medium Box        | 0.6                 | 7/18/07   |
| 32       | 7/16/07    | 2-High          | Medium Box        | 0.65                | 7/18/07   |
| 35       | 10/23/07   | 4-Not Specified | Wrap Bag          | 0.52                | 10/24/07  |
| 35       | 10/23/07   | 4-Not Specified | Small Box         | 0.58                | 10/25/07  |
| 36       | 11/3/07    | 1-Urgent        | Small Box         | 0.55                | 11/3/07   |
| 65       | 3/18/07    | 1-Urgent        | Small Pack        | 0.49                | 3/19/07   |
| 66       | 1/20/05    | 5-Low           | Wrap Bag          | 0.56                | 1/20/05   |

*Example Dataset of Orders with Attribute Types*

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# *Data Semantics*

## *Abstract Semantics*

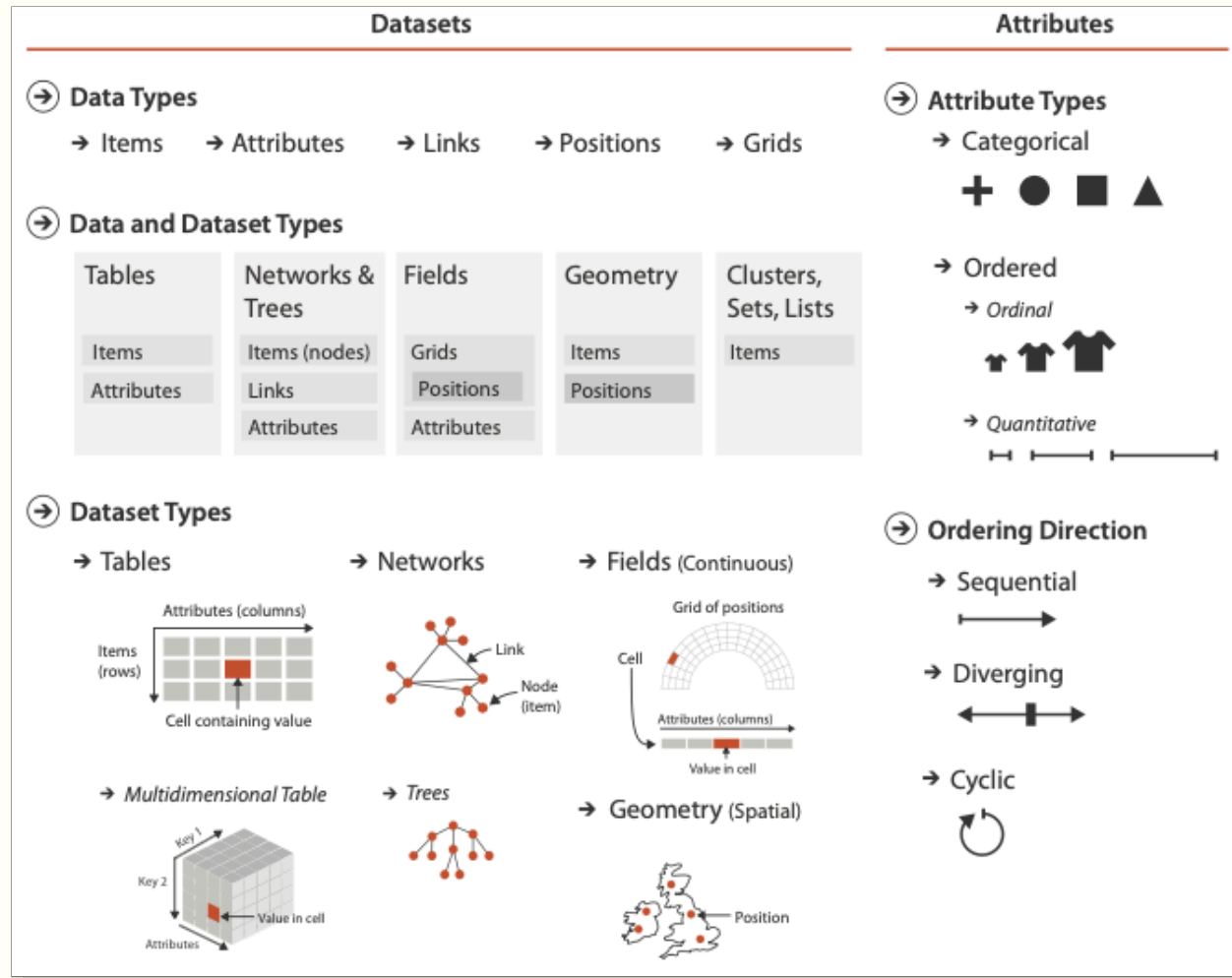
1. Key / Index
2. Value / Measure

## *Abstract Semantics*

1. Key / Index
2. Value / Measure

## *Structural Semantics*

1. Spatial
2. Temporal
3. Field
4. Vector



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*Determine the **Semantics** of each variable below as either a **Key** or **Value**, then add any relevant structural semantics (Spatial / Temporal / Field / Vector) if one applies:*

### *Variables*

- 1. Student ID*
- 2. Exam Score*
- 3. Timestamp of Submission*
- 4. Zip Code*

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# *Data Transformations and Tidy Data*

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*Data is rarely ready to visualize right away,  
you'll often need to **transform**, **merge**, or **tidy** it  
first.*

# The Relational Model

## *Relational Data*

1. *Table*: Dataset
2. *Row*: Record
3. *Column*: variable (name + type)
4. *Schema*: Table's column names & types
5. *Database*: Set of tables

| <i>Team</i> | <i>Wins</i> | <i>Losses</i> |
|-------------|-------------|---------------|
| Warriors    | 10          | 4             |
| Kings       | 7           | 7             |
| Lakers      | 3           | 11            |

Each *row* is a record, each *column* is a variable, each *cell* is a value.

# Relational Algebra

Table(s) in, table out.

## *Core Operations*

1. *select* — choose a set of columns
2. *filter* — remove unwanted rows
3. *order by* — sort records by column values
4. *group by* + aggregate (sum, mean, ...) — partition, then summarize
5. *join / union* — combine multiple tables

Implemented by SQL, pandas, polars, *arquero*, and similar table libraries.

# Select & Filter

| <i>A</i> | <i>B</i> | <i>C</i> |
|----------|----------|----------|
| x        | 1        | 4        |
| y        | 2        | 5        |
| z        | 3        | 6        |

select "A", "B"

| <i>A</i> | <i>B</i> |
|----------|----------|
| x        | 1        |
| y        | 2        |
| z        | 3        |

filter "B" % 2 = 1

| <i>A</i> | <i>B</i> | <i>C</i> |
|----------|----------|----------|
| x        | 1        | 4        |
| z        | 3        | 6        |

# Group By + Aggregate

group by "A"; aggregate mean

| Original |          |          | → Aggregated |          |          |
|----------|----------|----------|--------------|----------|----------|
| <i>A</i> | <i>B</i> | <i>C</i> | <i>A</i>     | <i>B</i> | <i>C</i> |
| x        | 1        | 4        | x            | 5.33     | 7.67     |
| y        | 2        | 5        | y            | 6.5      | 8.5      |
| z        | 3        | 6        | z            | 3        | 6        |
| x        | 7        | 9        |              |          |          |
| x        | 8        | 10       |              |          |          |
| y        | 11       | 12       |              |          |          |

# Combining Tables: Join

inner join  $T_1, T_2$  on “A”

| $T_1$    |          |          | $T_2$    |          |          | $\longrightarrow$ |  |  |  |  |  | Result   |          |          |          |          |  |
|----------|----------|----------|----------|----------|----------|-------------------|--|--|--|--|--|----------|----------|----------|----------|----------|--|
| <i>A</i> | <i>B</i> | <i>C</i> | <i>A</i> | <i>D</i> | <i>E</i> |                   |  |  |  |  |  | <i>A</i> | <i>B</i> | <i>C</i> | <i>D</i> | <i>E</i> |  |
| x        | 1        | 4        | x        | 7        | 10       |                   |  |  |  |  |  | x        | 1        | 4        | 7        | 10       |  |
| y        | 2        | 5        | why      | 8        | 11       |                   |  |  |  |  |  | z        | 3        | 6        | 9        | 12       |  |
| z        | 3        | 6        | z        | 9        | 12       |                   |  |  |  |  |  |          |          |          |          |          |  |

An *inner join* keeps only matching keys. An *outer join* keeps every key filling in missing values.

Based on this, what would a *left join* keep?

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*Tidy Data*

# Tidy Data

## *Rules of Tidy Data*

1. Each *observation* is a row
2. Each *variable* is a column
3. Each *value* is a cell

| <i>month</i> | <i>temp</i> | <i>precip</i> | <i>wind</i> |
|--------------|-------------|---------------|-------------|
| January      | 56.55       | 0.06          | 5.32        |
| February     | 56.10       | 0.12          | 5.46        |
| March        | 56.36       | 0.11          | 6.52        |

Tidy data relies on *both* a conceptual model (what's counts as an observation?) and a data model.

# Wide vs Narrow Tables

Better for *data entry* vs *data analysis*

*Tidy*

| <i>month</i> | <i>temp</i> | <i>precip</i> | <i>wind</i> |
|--------------|-------------|---------------|-------------|
| January      | 56.55       | 0.06          | 5.32        |
| February     | 56.10       | 0.12          | 5.46        |

*Untidy*

| <i>month</i> | <i>element</i> | <i>value</i> |
|--------------|----------------|--------------|
| January      | temp           | 56.55        |
| January      | precip         | 0.06         |
| January      | wind           | 5.32         |
| February     | temp           | 56.10        |
| February     | precip         | 0.12         |
| February     | wind           | 5.46         |

In the untidy table, a single observation is spread across several rows, and the *value* column mixes different variables. Reshaping between these forms is called *pivoting* and *unpivoting*.

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## Discussion

*Is this table tidy? What are its variables and its observations?*

| <i>artist</i>       | <i>track</i>                  | <i>date_entered</i> | <i>wk1</i> | <i>wk2</i> | <i>wk3</i> |
|---------------------|-------------------------------|---------------------|------------|------------|------------|
| <i>2 Pac</i>        | <i>Baby Don't Cry</i>         | <i>2000-02-26</i>   | <i>87</i>  | <i>82</i>  | <i>72</i>  |
| <i>3 Doors Down</i> | <i>Kryptonite</i>             | <i>2000-04-08</i>   | <i>81</i>  | <i>70</i>  | <i>68</i>  |
| <i>98°</i>          | <i>Give Me Just One Night</i> | <i>2000-08-19</i>   | <i>51</i>  | <i>39</i>  | <i>34</i>  |

*Weekly Billboard ranks, 2000 · R for Data Science, Hadley Wickham*

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*One person's tidy is another person's mess*

*This means that most real analyses will require at least a little tidying. You'll begin by figuring out what the underlying **variables** and **observations** are. Sometimes this is easy; other times you'll need to consult with the people who originally generated the data.*

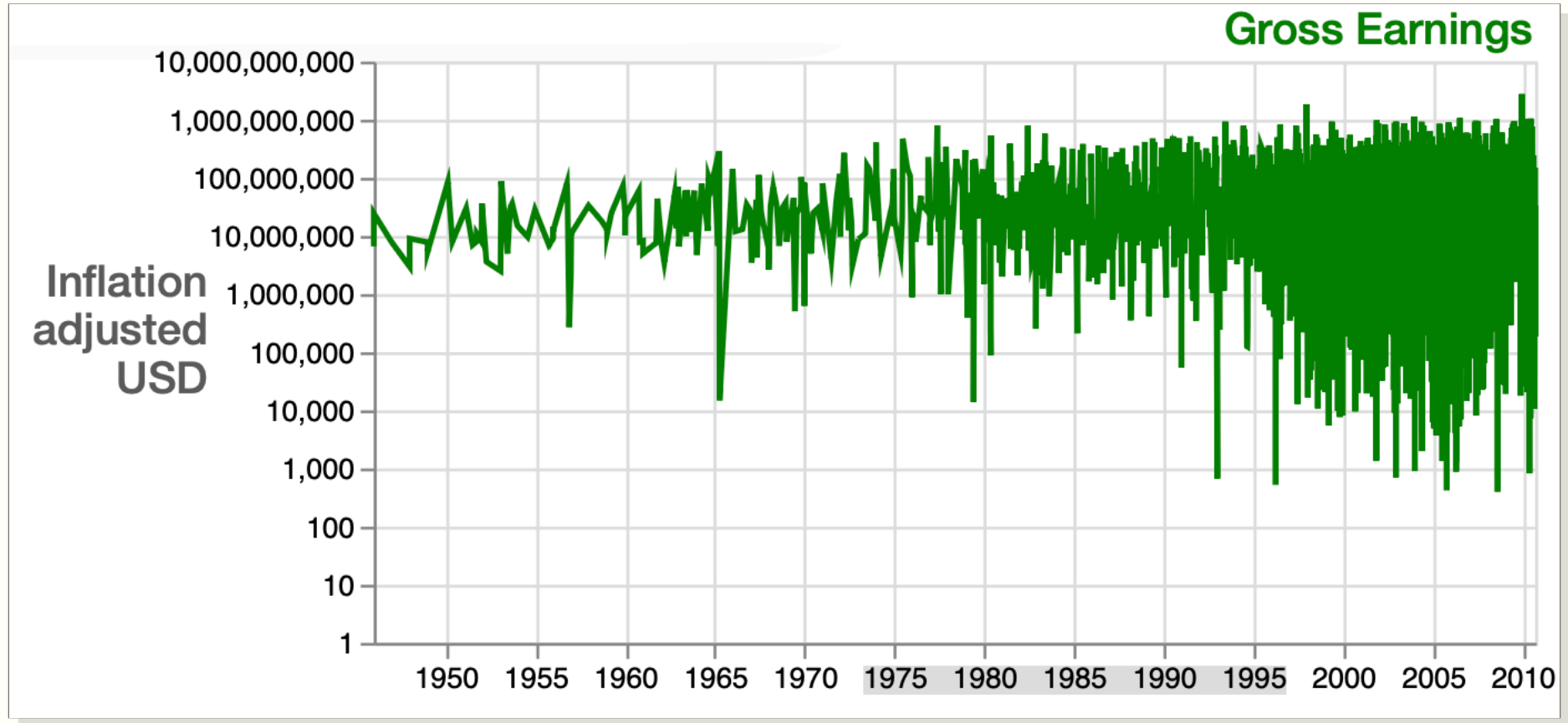
*— Hadley Wickham*

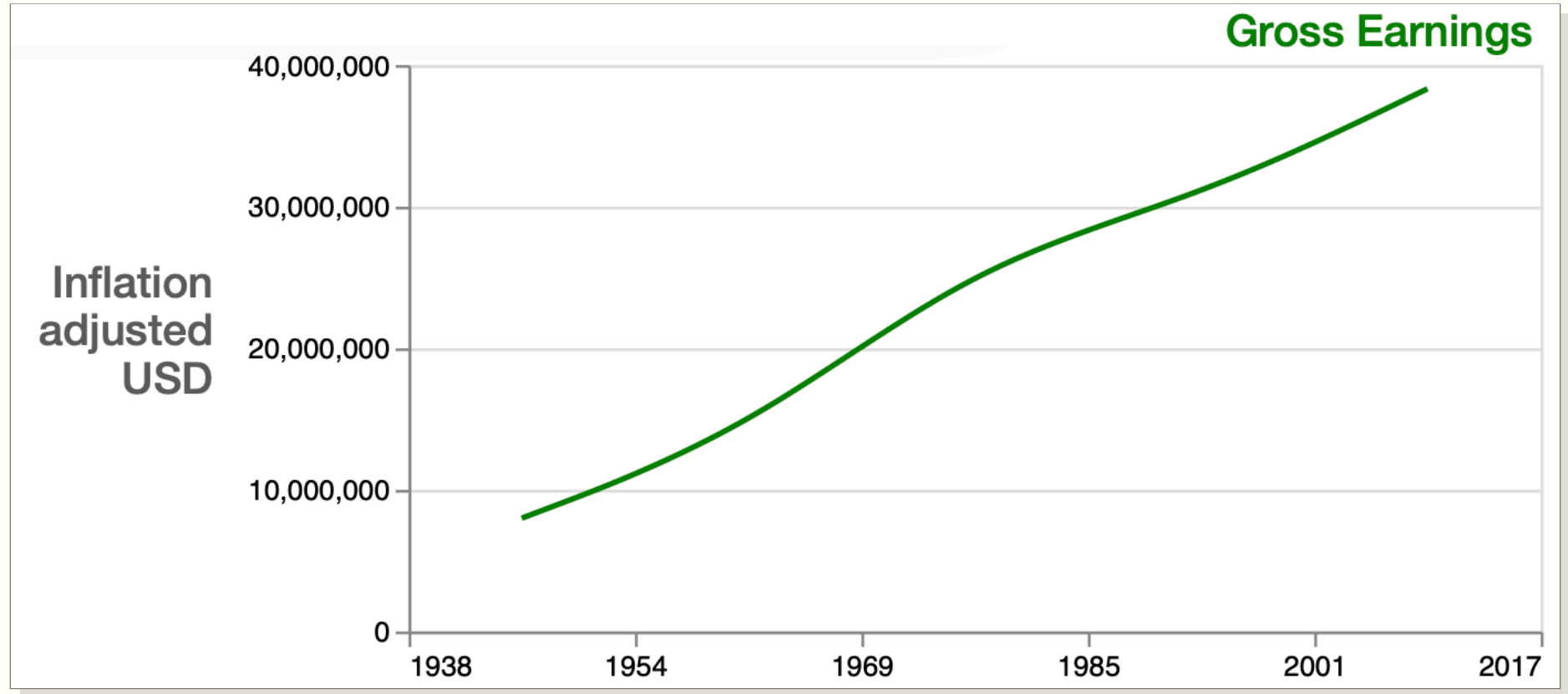
*R for Data Science*, “Data Tidying”

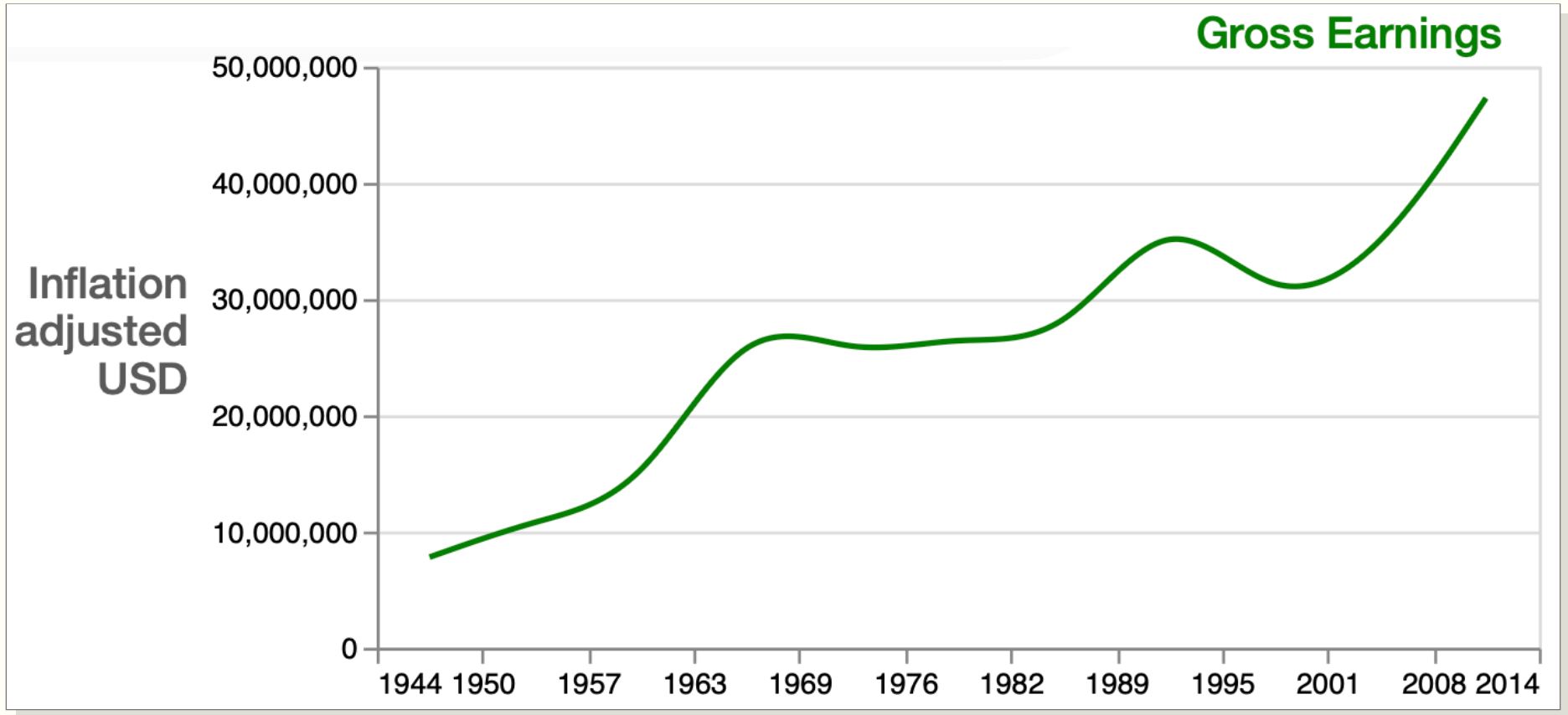


*Visualization (In)Dependent Transformations*

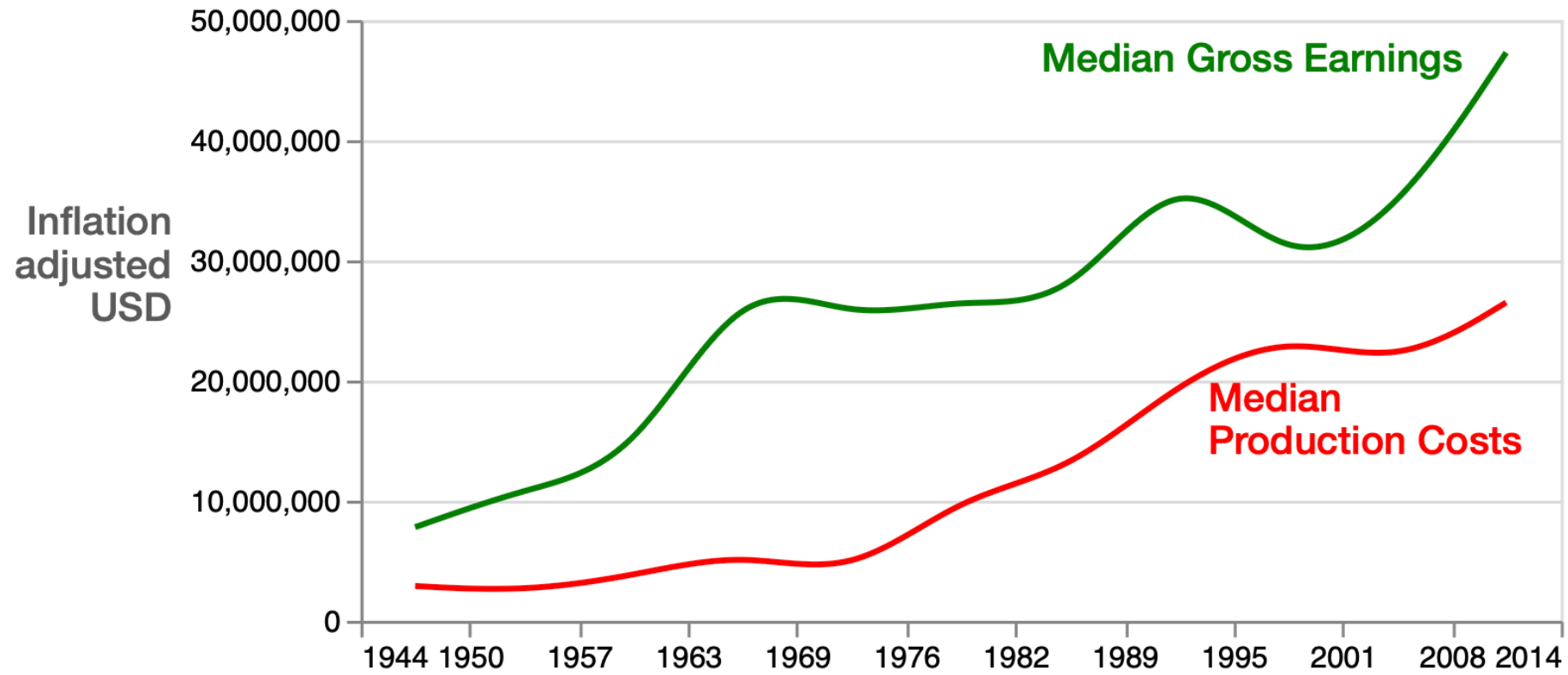
| Title                      | Release_Date | Worldwide_Gross | Production_Budget | IMDB_Rating |
|----------------------------|--------------|-----------------|-------------------|-------------|
| The Land Girls             | Jun 12 1998  | 146083.0        | 8000000.0         | 6.1         |
| First Love, Last Rites     | Aug 07 1998  | 10876.0         | 300000.0          | 6.9         |
| I Married a Strange Person | Aug 28 1998  | 203134.0        | 250000.0          | 6.8         |
| Let's Talk About Sex       | Sep 11 1998  | 373615.0        | 300000.0          | NaN         |
| Slam                       | Oct 09 1998  | 1087521.0       | 1000000.0         | 3.4         |
| ...                        | ...          | ...             | ...               | ...         |
| Zack and Miri Make a Porno | Oct 31 2008  | 36851125.0      | 24000000.0        | 7.0         |
| Zodiac                     | Mar 02 2007  | 83080084.0      | 85000000.0        | NaN         |
| Zoom                       | Aug 11 2006  | 12506188.0      | 35000000.0        | 3.4         |
| The Legend of Zorro        | Oct 28 2005  | 141475336.0     | 80000000.0        | 5.7         |
| The Mask of Zorro          | Jul 17 1998  | 233700000.0     | 65000000.0        | 6.7         |



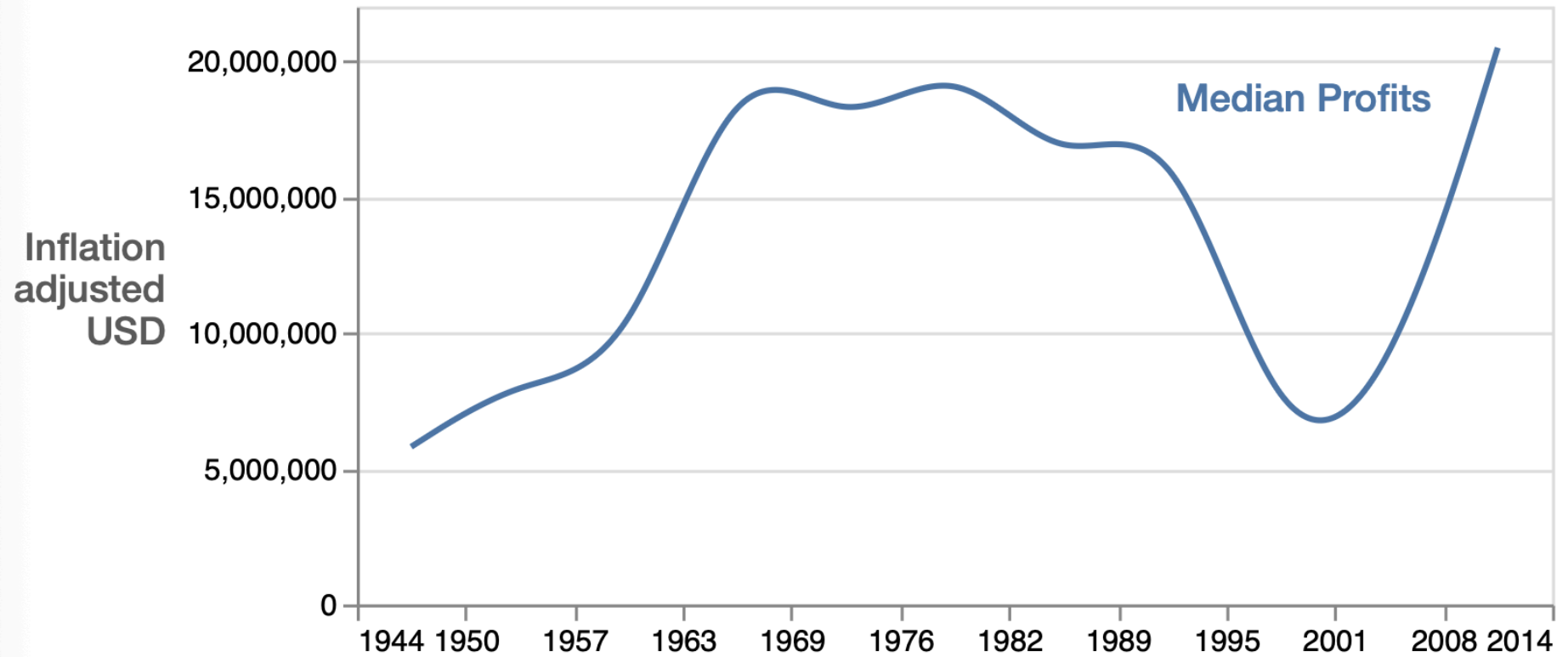




# Movie Business Growing Over Time



## Movie Business Growing Over Time?





# *Task Abstraction*

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*What does the **user actually do** with the  
visualization?*

# *User Actions*

## *Analyze*

1. Consume
  1. Discover
  2. Present
  3. Enjoy

# *User Actions*

## *Analyze*

1. Consume
  1. Discover
  2. Present
  3. Enjoy
2. Produce
  1. Annotate
  2. Record
  3. Derive

# *User Actions*

## *Analyze*

1. Consume
  1. Discover
  2. Present
  3. Enjoy
2. Produce
  1. Annotate
  2. Record
  3. Derive

## *Search*

1. Target Known
  1. Lookup
  2. Locate
2. Target Unknown
  1. Browse
  2. Explore

# *User Actions*

## *Analyze*

1. Consume
  1. Discover
  2. Present
  3. Enjoy
2. Produce
  1. Annotate
  2. Record
  3. Derive

## *Search*

1. Target Known
  1. Lookup
  2. Locate
2. Target Unknown
  1. Browse
  2. Explore

## *Query*

1. Identify
2. Compare
3. Summarize

---

## *Discussion*

*For each scenario, name the action (and sub-type) being performed:*

### *Scenarios*

- 1. Browsing the web for examples for Assignment 1*
- 2. A doctor looking for signs of cancer in an MRI scan*
- 3. Checking whether wildfire acreage this year exceeds last year's*
- 4. Checking if the error bars on a bar chart in a publication match the claims in the paper*
- 5. Screenshotting a chart from social media to put in a slide deck*

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*User **Actions** (verbs) vs User **Targets** (nouns)*

# *User Targets*

## *Full Dataset*

1. Trends
2. Features
3. Outliers / Anomalies

# *User Targets*

## *Full Dataset*

1. Trends
2. Features
3. Outliers / Anomalies

## *Attributes*

1. Single Attribute
  1. Distribution
  2. Paradigms
  3. Extremes
2. Multiple Attributes
  1. Dependency
  2. Correlation
  3. Similarity

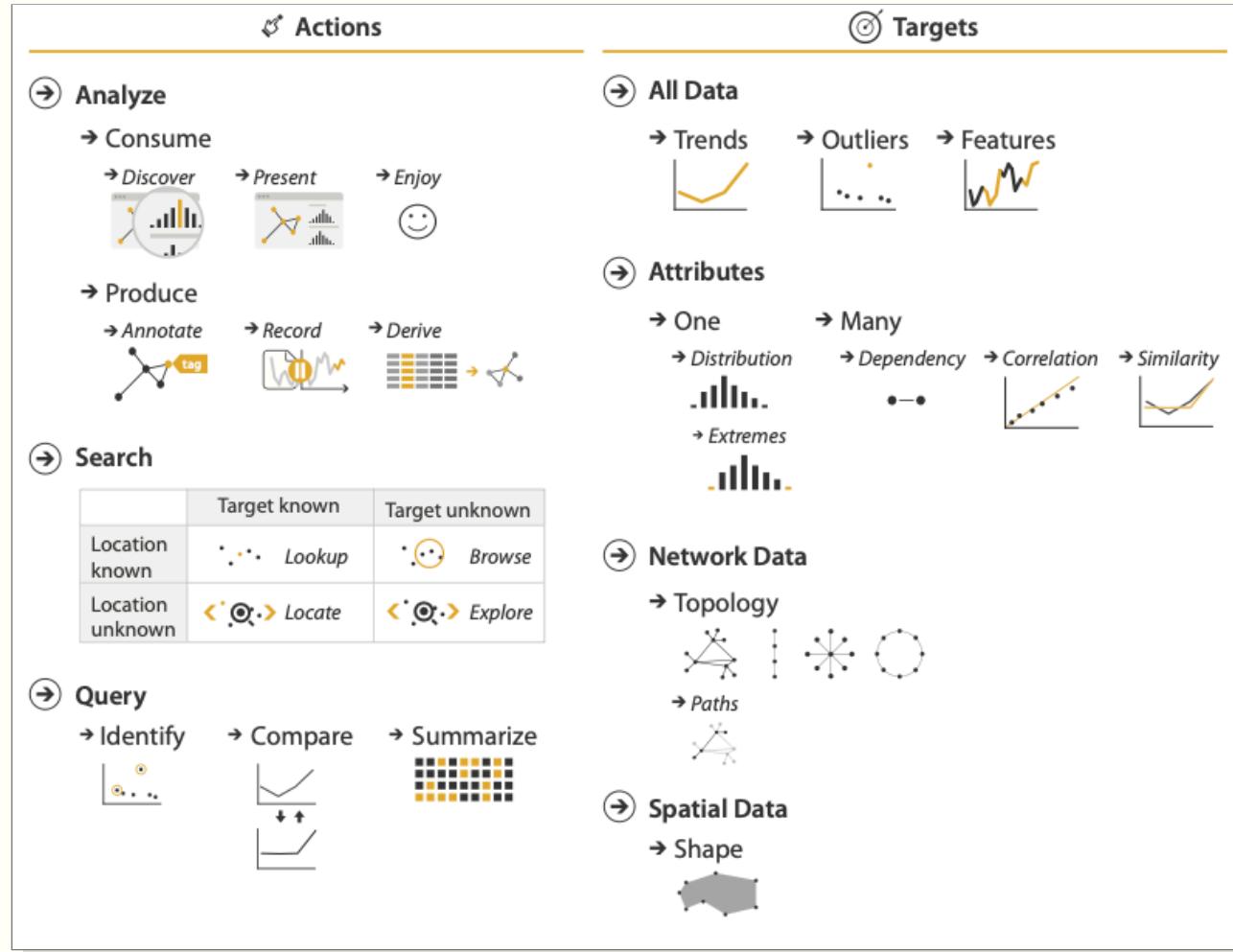
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## Discussion

*Imagine you are looking at a scatterplot of height vs. weight for your class. Which **user target** does each question create?*

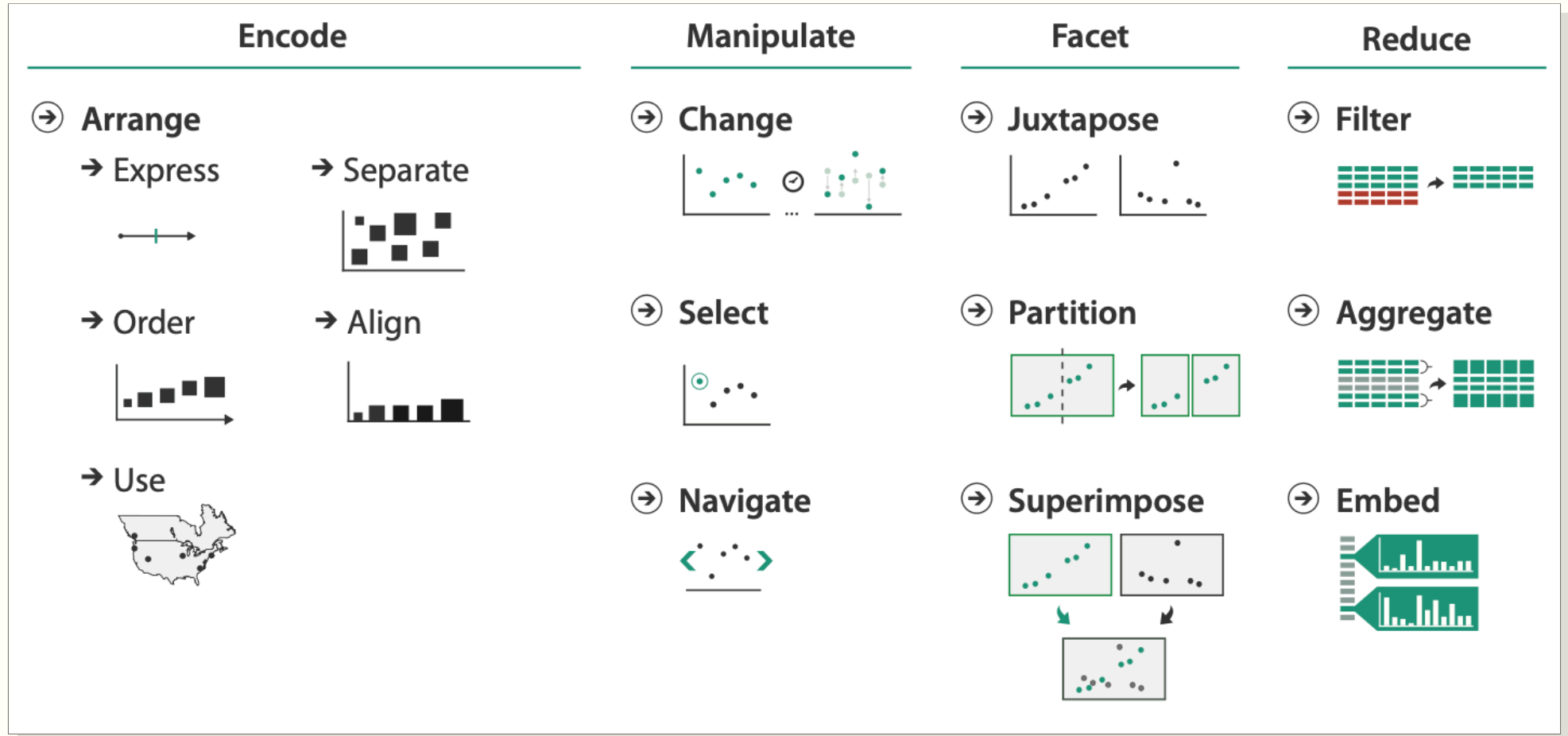
### Questions

1. *“Is there a relationship between height and weight?”*
2. *“Who is the tallest person in the room?”*
3. *“Are heights roughly bell-shaped, or skewed?”*
4. *“Are there any distinct clusters?”*



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*How do designs support **user tasks**?*

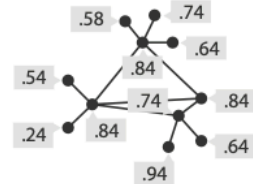


Munzner 2014, Figure 3.7

### Task 1



**In**  
Tree



**Out**  
Quantitative  
attribute on nodes



**What?**

- **In** Tree
- **Out** Quantitative attribute on nodes

**Why?**

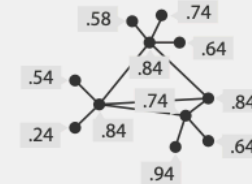
- Derive

### Task 2



**In**  
Tree

+



**In**  
Quantitative  
attribute on nodes



**Out**  
Filtered tree  
Removed  
unimportant parts

**What?**

- **In** Tree
- **In** Quantitative attribute on nodes
- **Out** Filtered tree

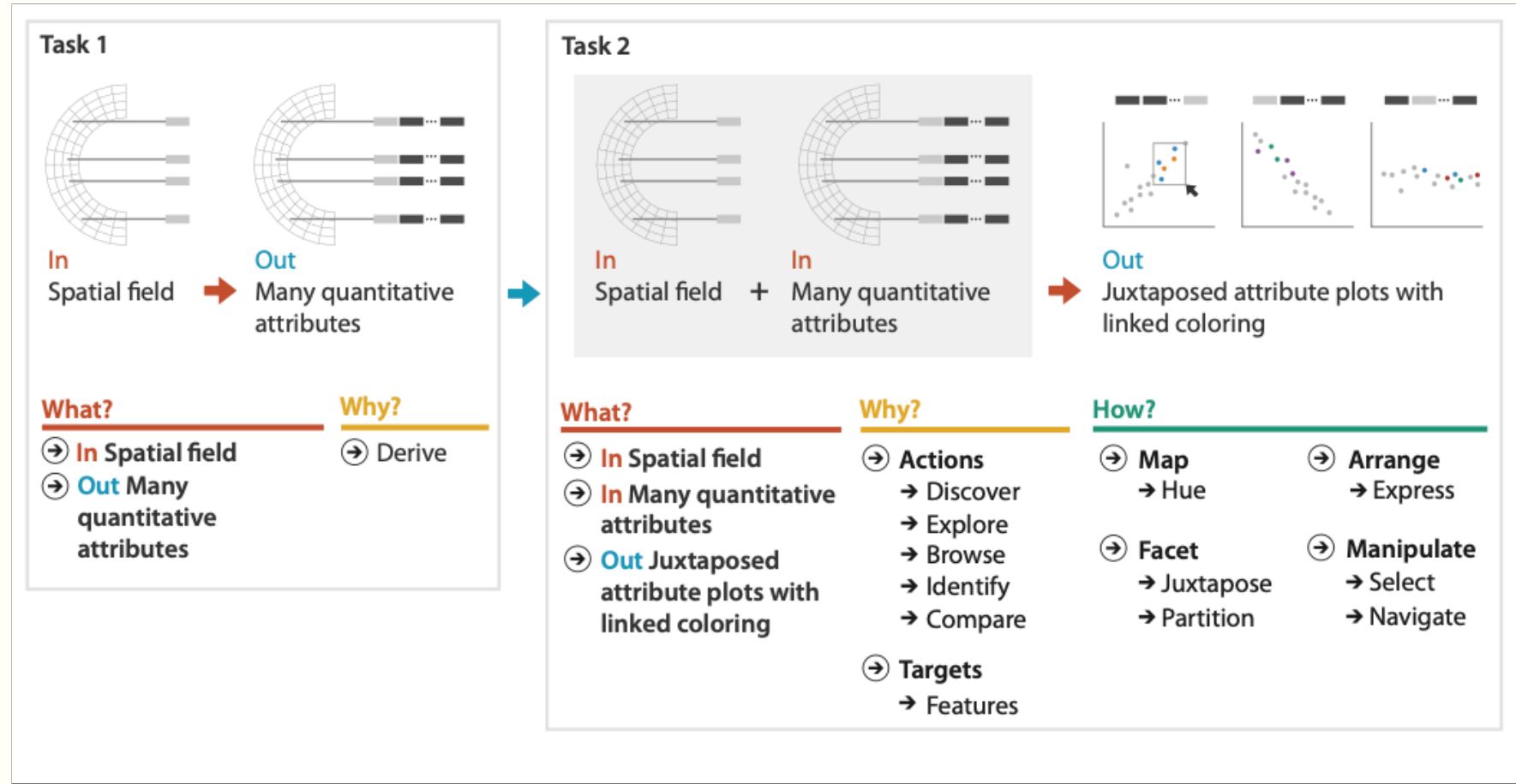
**Why?**

- Summarize
- Topology

**How?**

- Reduce
- Filter

Munzner 2014, Figure 3.11



Munzner 2014, Figure 3.13

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*How do we **actually make** good designs?*

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# *Sketching and the Design Process*

*Based on slides from Tal Wolman*

*The design space of possible visualizations is huge, and includes considerations of both **how to create** and **how to interact** with visual representations. Vis design is full of trade-offs, and most possibilities in the design space **are ineffective** for a particular task — designers must weigh the resource limitations of computers, of humans, and of displays.*

*— Tamara Munzner*

Visualization Analysis & Design, 2014

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*Every task has countless possible designs, most of them are bad, and you can't tell which from a distance.*

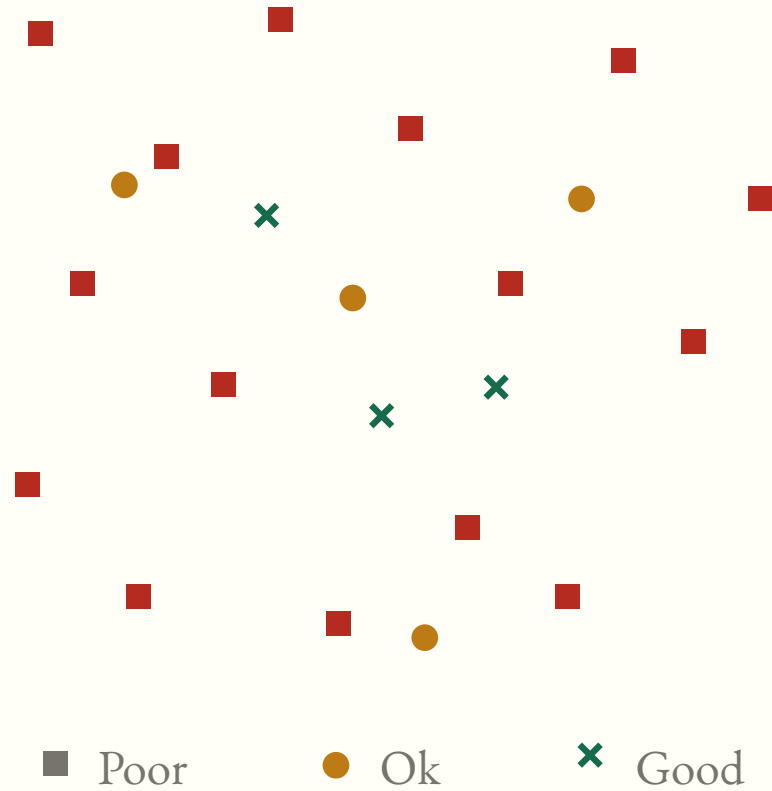
# Space of Possible Solutions



# Space of Possible Solutions



# Space of Possible Solutions

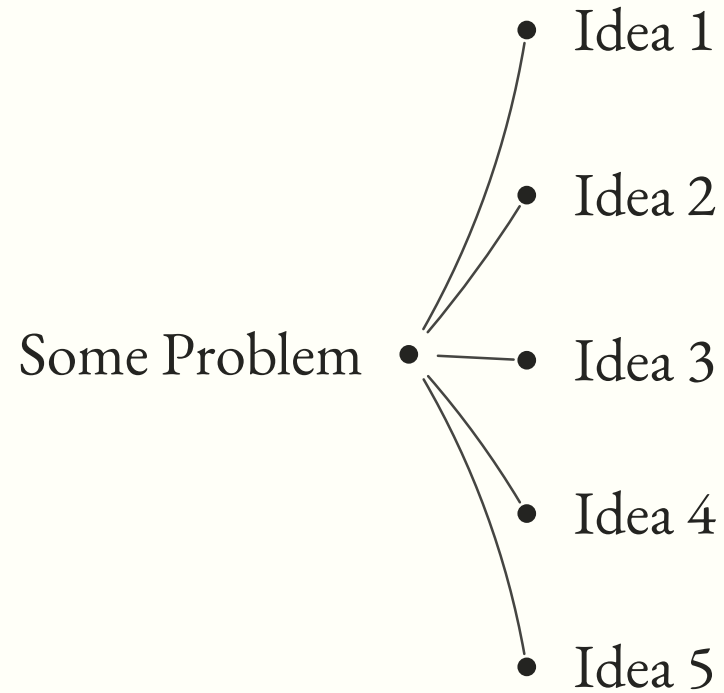


*The problem of a small consideration space is the higher probability of settling for an OK solution and missing a good one. A fundamental principle of design is to **consider multiple alternatives** and then choose the best, rather than fixating on the first idea. One way to guarantee this: **explicitly generate multiple ideas** in parallel.*

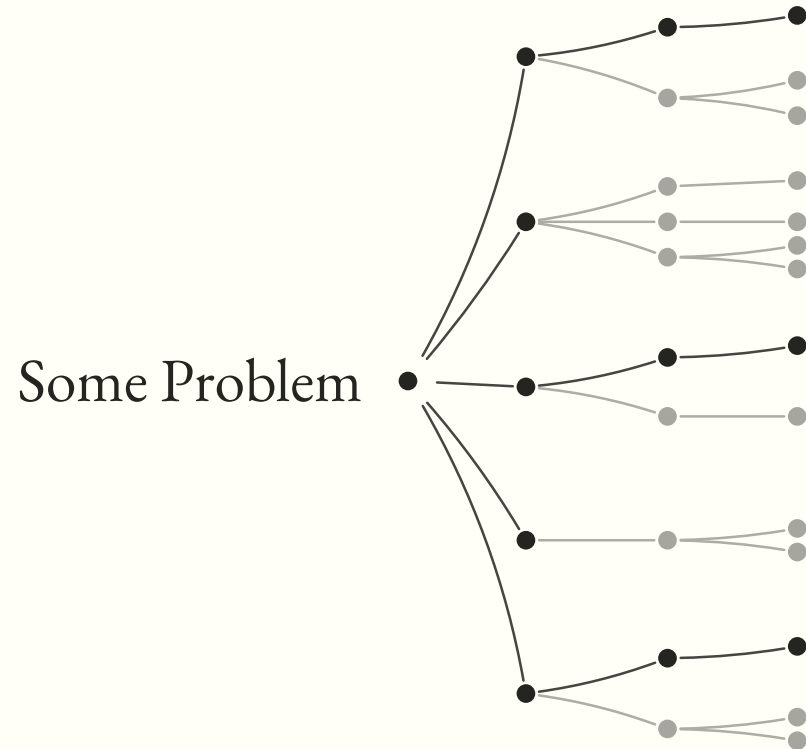
— *Tamara Munzner*

Visualization Analysis & Design, 2014

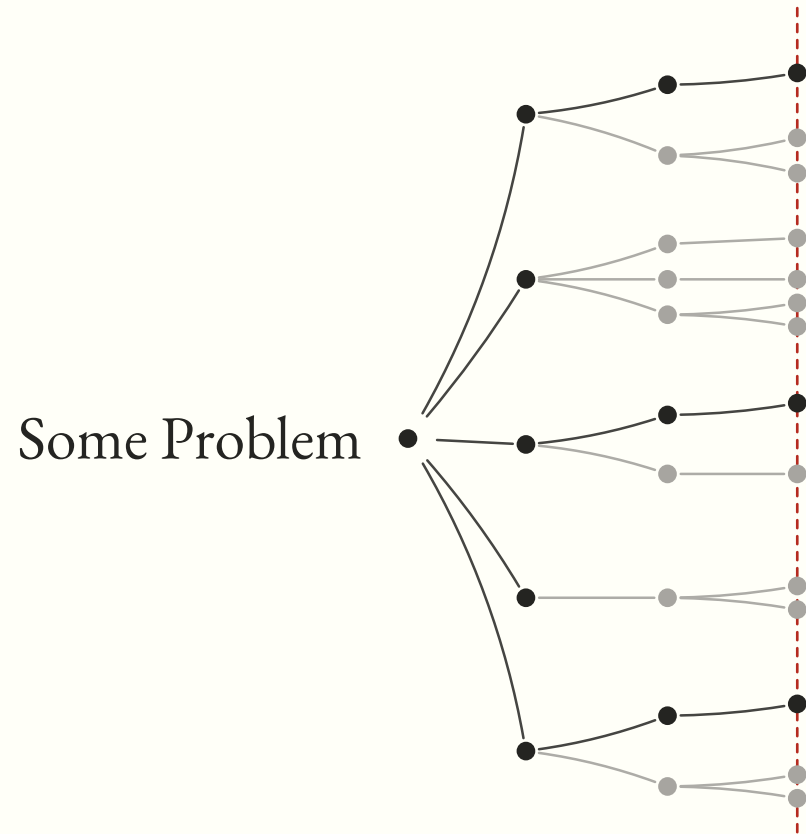
# Ideation as a Tree



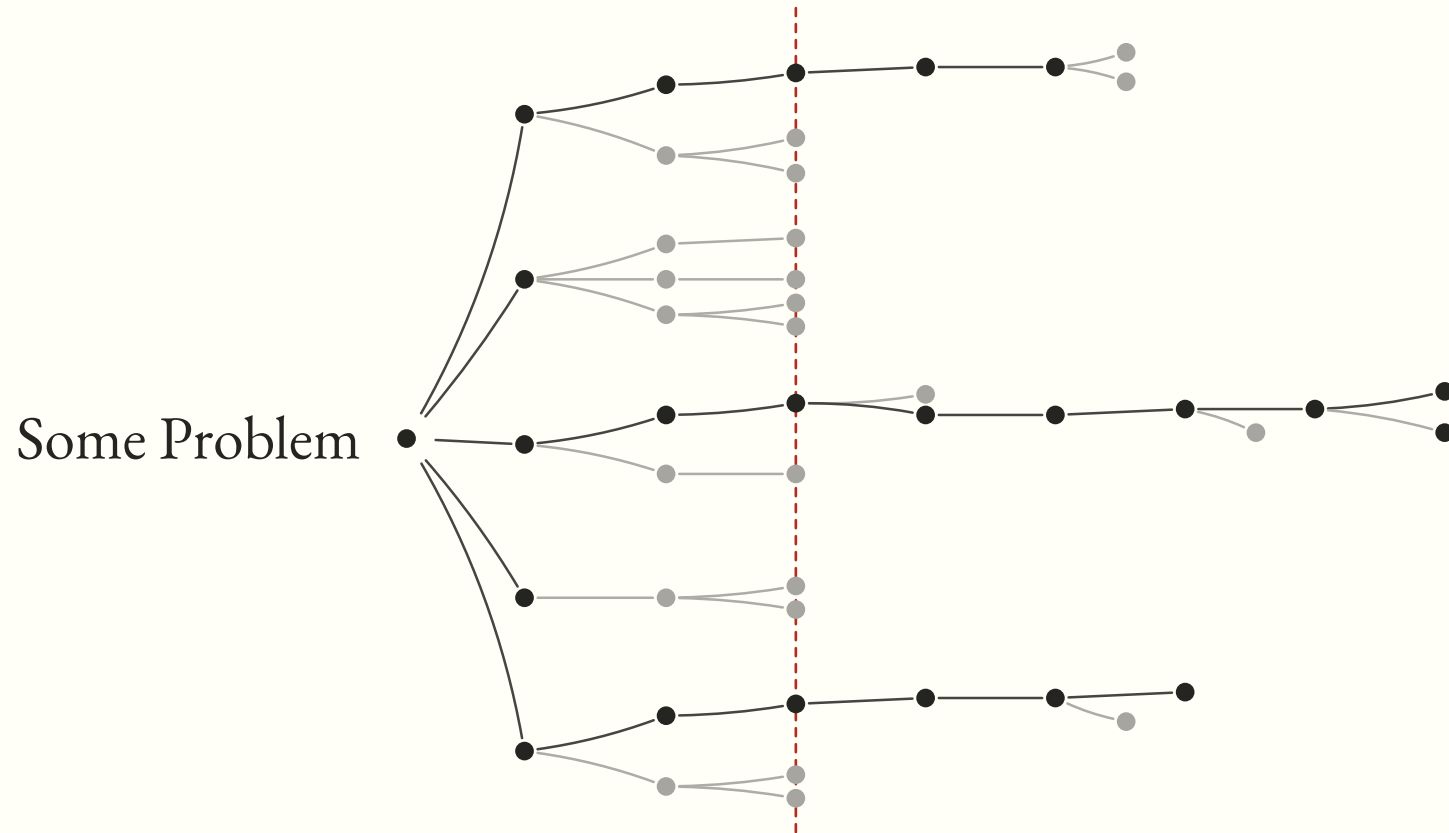
# Ideation as a Tree



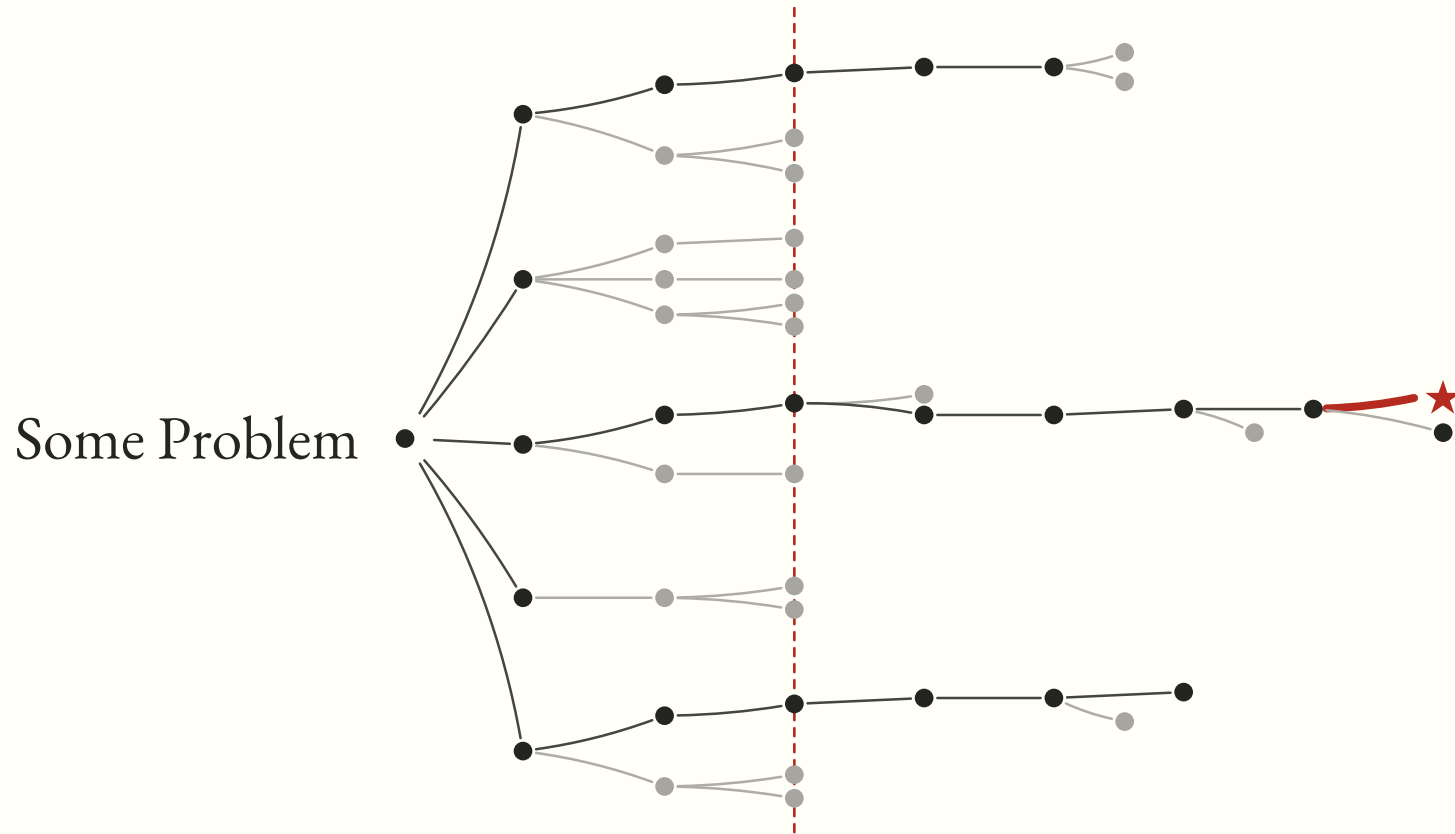
# Ideation as a Tree



# Ideation as a Tree



# Ideation as a Tree

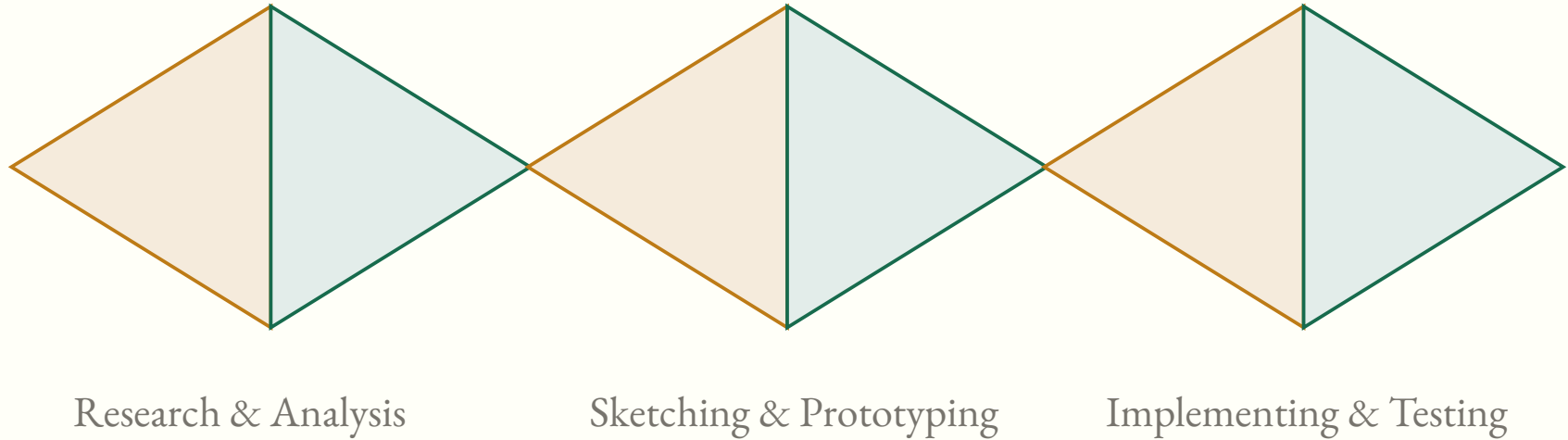


| *The best way to have a good idea is to **have lots of ideas.***

— *Linus Pauling*

Professor of Chemistry, Caltech · UC San Diego · Stanford — only person awarded two unshared Nobel Prizes

# Elaborate and Reduce



*Sketching is fundamental to ideation and design.*

*Traditional disciplines — industrial design, graphic design, architecture — make extensive use of sketches to develop, explore, communicate, and evaluate ideas.*

— *Tobidi, Buxton, Baecker & Sellen*

*User Sketches: A Quick, Inexpensive, and Effective Way to Elicit More Reflective User Feedback*, NordiCHI 2006

*Sketching has long been a **best practice** for designers. Through sketches, designers follow a generative process of **developing, honing, and choosing ideas**. Designers also use sketches to discuss, exchange, and critique ideas with others.*

*— Greenberg, Carpendale, Marquardt & Buxton*

*Sketching User Experiences: The Workbook, 2012*

*Sketches allow for a **dialog between the sketch and the viewer** — even when the viewer is the sketcher themselves — that **facilitates better understanding** of the problem, and in turn, generation of new ideas.*

*— Tohidi, Buxton, Baecker & Sellen*

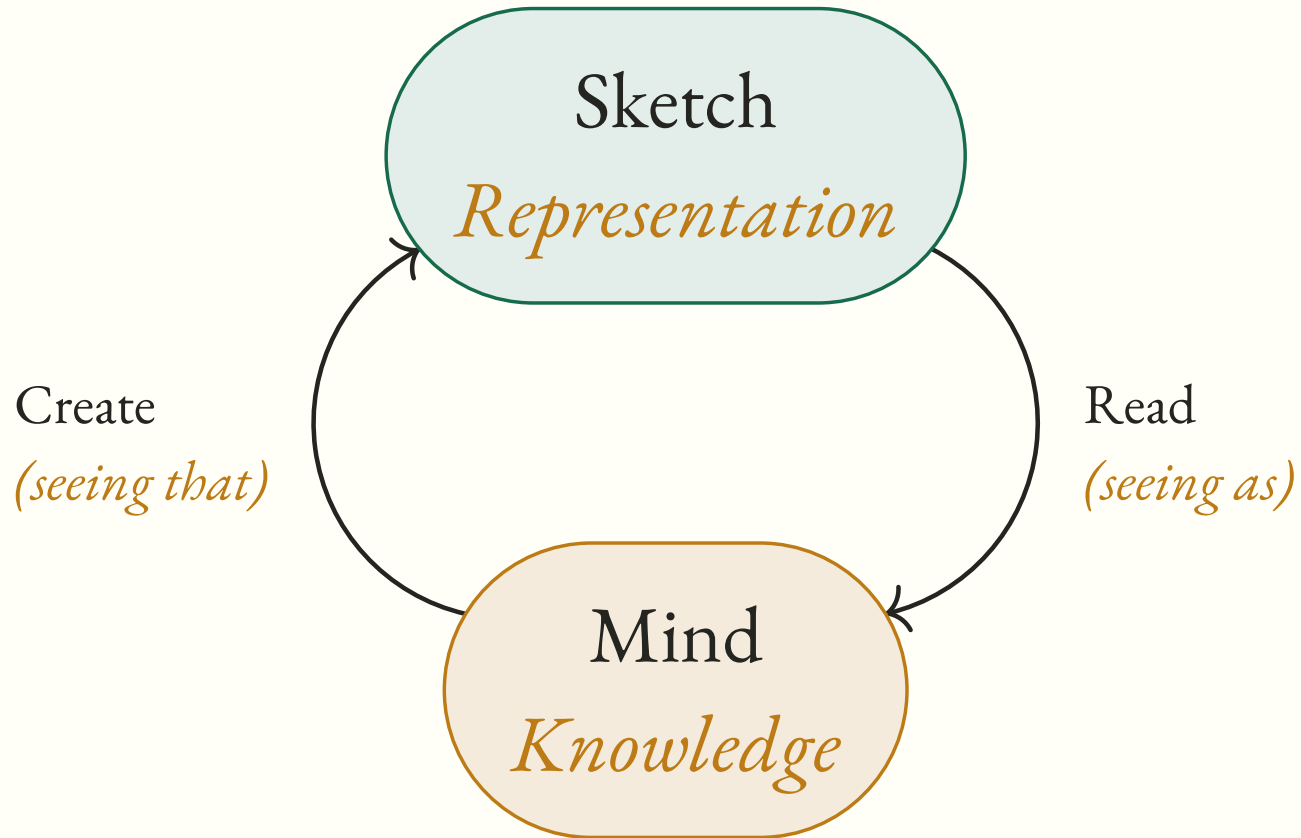
NordiCHI 2006

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*Sketching is **not about drawing!***

*Often the best sketches are just rough line drawings.*

# Sketching is a *Conversation with the Self*



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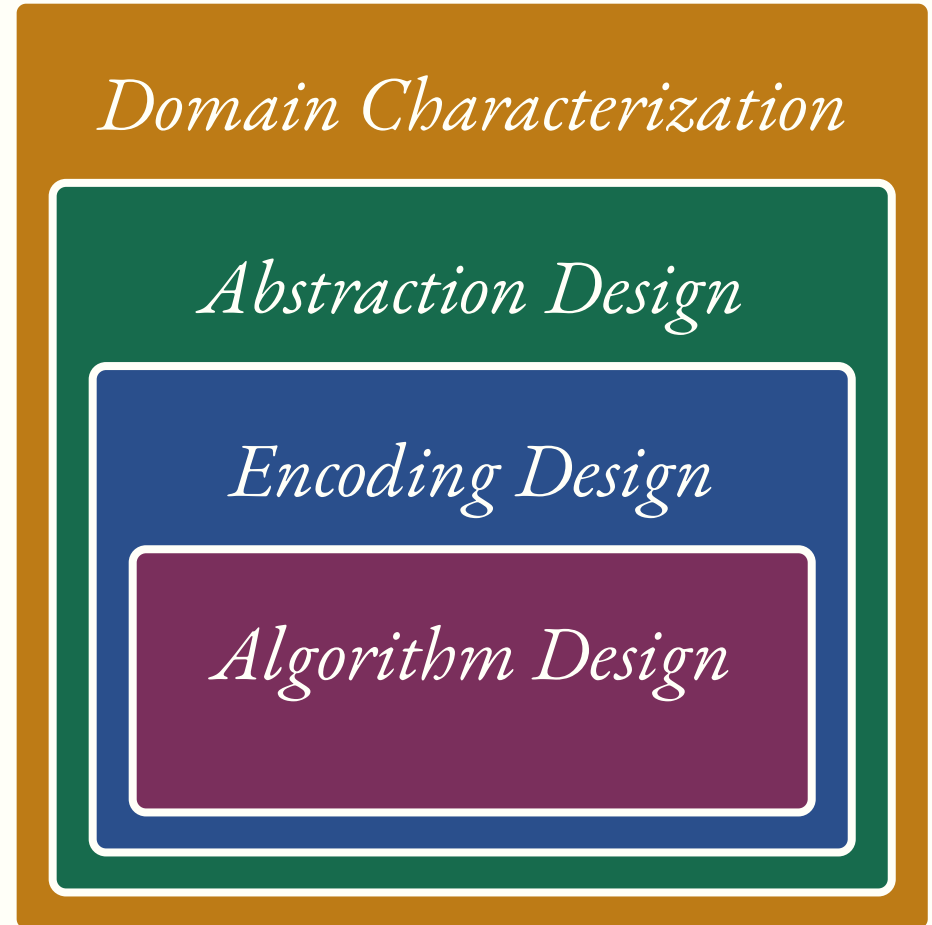
*Did we build the visualization right?*

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*Did we build the visualization right?*

*Did we build the **right** visualization?*

# The Nested Model



## *Domain Characterization*

*Who are the target users?*

*What are their needs?*

*Observe and interview*

## *Abstraction Design*

## *Encoding Design*

## *Algorithm Design*

## *Domain Characterization*

### *Abstraction Design*

*Data abstraction: what is being shown?*

*Task abstraction: why are they looking at it?*

### *Encoding Design*

### *Algorithm Design*

# *Domain Characterization*

## *Abstraction Design*

### *Encoding Design*

*Idioms: how is it shown?*

*How is it displayed?*

*How is it manipulated and interacted with?*

### *Algorithm Design*

*Domain Characterization*

*Abstraction Design*

*Encoding Design*

*Algorithm Design*

*How is it computed efficiently?*

*How is it computed accurately?*

*Domain Characterization*

*Abstraction Design*

*Encoding Design*

*Algorithm Design*

*How is it computed efficiently?*

*How is it computed accurately?*

Implementation

# Different Ways to Get it Wrong



# Different Ways to Get it Wrong

*Domain Characterization*

*Solving the wrong problem*

*Abstraction Design*

*Encoding Design*

*Algorithm Design*

# Different Ways to Get it Wrong

*Domain Characterization*

*Abstraction Design*

*Abstractions don't address the problem*

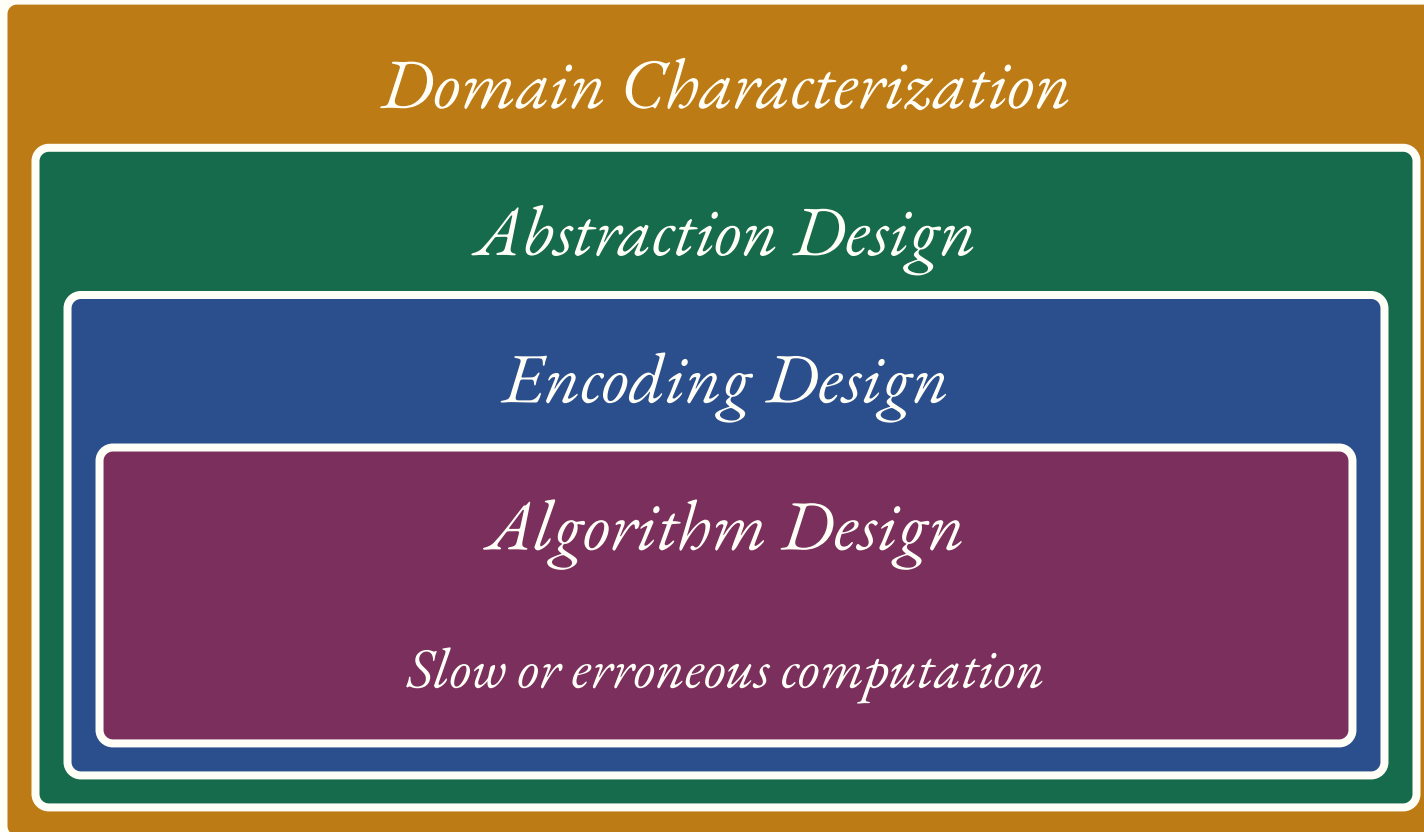
*Encoding Design*

*Algorithm Design*

# Different Ways to Get it Wrong



# Different Ways to Get it Wrong



*Domain Characterization*

*Abstraction Design*

*Encoding Design*

*Algorithm Design*

*Analyze computational complexity*

Implementation

*Measure system performance*

CS

## *Domain Characterization*

### *Abstraction Design*

#### *Encoding Design*

*Justify design choices*

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*Analyze results qualitatively*

*Measure human responses in lab*

*General usability tests*

Design  
CS  
Psych  
HCI

## *Domain Characterization*

*Observe target users with existing tools*

### *Abstraction Design*

*Observe use of tool in context*

*Target users' usability tests*

*Measure adoption and impact*

Social Sciences

Design

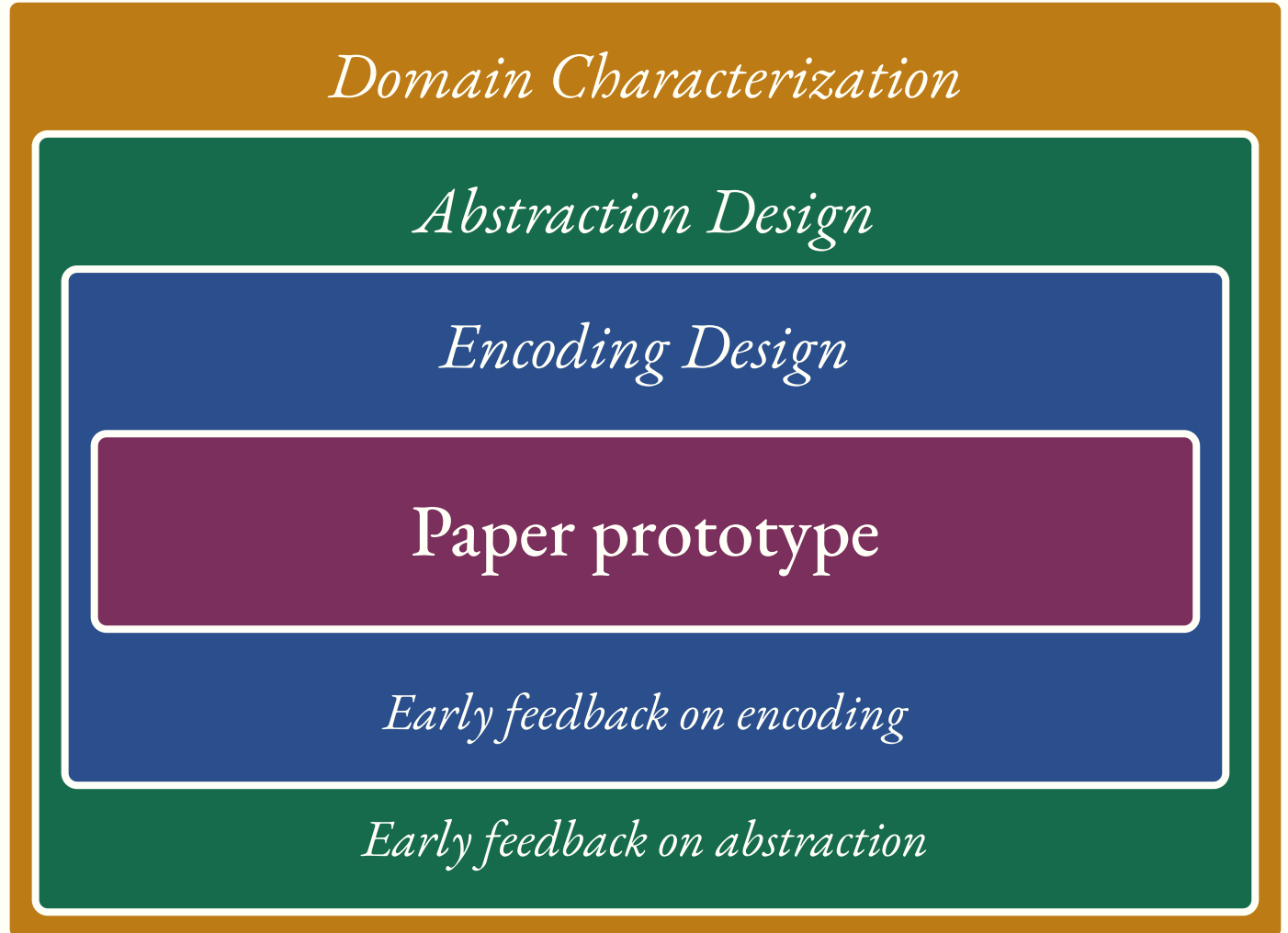
CS

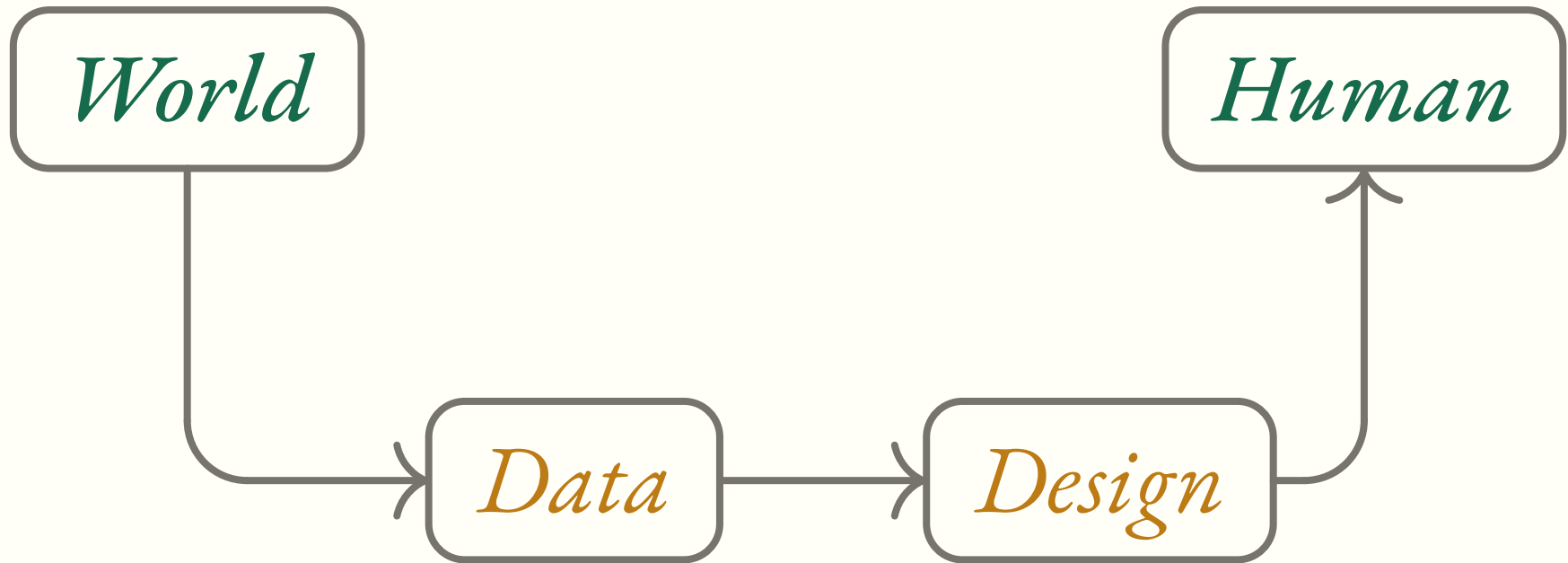
Psych

HCI

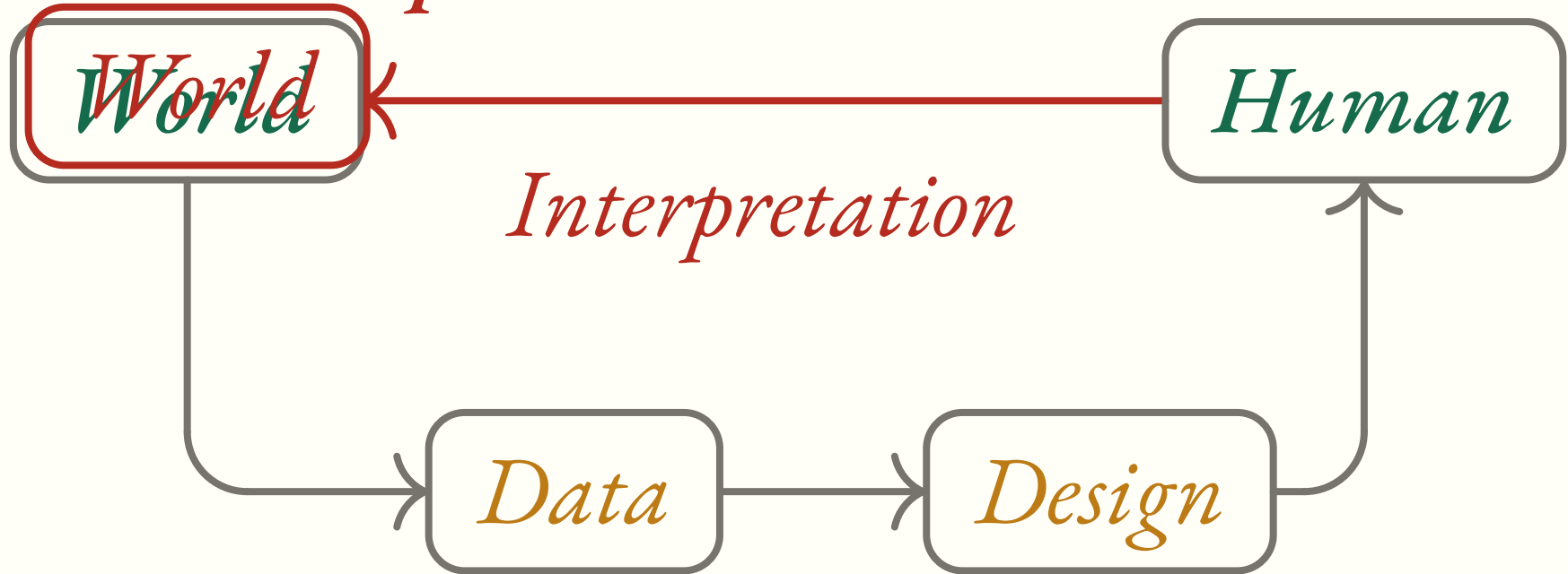
Social Sciences

Abstract the  
design tasks





*Mind the Gap!*



## *Conceptual Model*

